

EDS Theory Assignment 2

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Problem Statement:

A dataset (mail_data.csv) containing SMS/Email messages labelled as **spam** or **ham** (the Enron Email / SMS Spam Collection Dataset) has been provided. Each record contains a **Category** (spam/ham) and the raw **Message** text. You are required to analyze this dataset using **NumPy** for numerical computations and **Pandas** for data manipulation. Derived features including *msg_length*, *word_count*, *char_per_word*, and a binary *label* column (spam=1, ham=0) are engineered before analysis. Based on the dataset, formulate and solve **20 problem statements (10 NumPy + 10 Pandas)**.

NumPy Operations:

1. Display the first 10 message lengths as a NumPy array.
2. Find the total sum of all message lengths.
3. Compute the average message length across all messages.
4. Identify the maximum message length in the dataset.
5. Identify the minimum message length in the dataset.
6. Calculate the median word count across all messages.
7. Compute the standard deviation of message lengths.
8. Find the total number of spam messages (label = 1).
9. Sort all message lengths in descending order and show the top 10.
10. Find the index (position) of the longest message in the dataset.

Pandas Operations:

11. Display the first 5 rows of the dataset.
12. Display the last 5 rows of the dataset.
13. Show the column names and data types of the dataset.
14. Find the total number of rows and columns in the dataset.
15. Calculate the average message length per Category (spam/ham).
16. Find the total number of messages per Category.
17. Find the average word count per Category.
18. Identify the top 5 messages with the highest message length.
19. Identify the bottom 5 messages with the lowest message length.

20. Display summary statistics for key numeric columns.

Complete Code:

```

import numpy as np
import pandas as pd

np.set_printoptions(suppress=True)

# Load Dataset
df = pd.read_csv('mail_data.csv')

# Engineer derived features
df['label'] = (df['Category'] == 'spam').astype(int)
df['msg_length'] = df['Message'].str.len()
df['word_count'] = df['Message'].str.split().str.len()
df['char_per_word'] = (df['msg_length'] / df['word_count']).round(2)

# Convert to NumPy arrays
msg_len = df['msg_length'].to_numpy()
word_cnt = df['word_count'].to_numpy()
labels = df['label'].to_numpy()

# NUMPY OPERATIONS # 1. First 10
message_lengths print('1. First 10
Message Lengths:')
print(msg_len[:10])

# 2. Total sum
print('2. Total Sum of Message Lengths:')
print(np.sum(msg_len))

# 3. Average
print('3. Average Message Length:')
print(round(np.mean(msg_len), 2))

# 4. Maximum
print('4. Maximum Message Length:')
print(np.max(msg_len))

# 5. Minimum
print('5. Minimum Message Length:')
print(np.min(msg_len))

# 6. Median word count
print('6. Median Word Count:')
print(np.median(word_cnt))

# 7. Standard deviation
print('7. Std Dev of Message Lengths:')
print(round(np.std(msg_len), 4))

# 8. Spam count
print('8. Number of Spam Messages (label=1):')
print(np.sum(labels == 1))

# 9. Top 10 descending
print('9. Top 10 Message Lengths (Descending):') print(np.sort(msg_len)[::-1][:10])

```

```
# 10. Index of longest
print('10. Index of the Longest Message:')
print(np.argmax(msg_len))

# PANDAS OPERATIONS # 11.
First 5 rows print('11.
First 5 rows:')
print(df[['Category', 'Message', 'msg_length', 'word_count']].head().to_string(index=False))

# 12. Last 5 rows
print('12. Last 5 rows:')
print(df[['Category', 'Message', 'msg_length', 'word_count']].tail().to_string(index=False))

# 13. Column info
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```
print('13. Column Info:')
print(df.dtypes.to_string())

# 14. Shape
print('14. Dataset Shape:')
print(df.shape)

# 15. Avg length by Category
print('15. Avg Message Length by Category:')
print(df.groupby('Category')['msg_length'].mean().round(2).to_string())

# 16. Count per Category
print('16. Messages per Category:')
print(df['Category'].value_counts().to_string())

# 17. Avg word count by Category
print('17. Avg Word Count by Category:')
print(df.groupby('Category')['word_count'].mean().round(2).to_string())

# 18. Top 5 longest
print('18. Top 5 Longest Messages:')
print(df.nlargest(5, 'msg_length')[['Category', 'msg_length', 'word_count']].to_string(index=False))

# 19. Bottom 5 shortest
print('19. Bottom 5 Shortest Messages:')
print(df.nsmallest(5, 'msg_length')[['Category', 'msg_length', 'word_count']].to_string(index=False))

# 20. Summary statistics
print('20. Summary Statistics:')
print(df[['msg_length', 'word_count', 'char_per_word', 'label']].describe().round(2).to_string())
```

Output Screenshots:

Output of NumPy Operations:

```
1. First 10 Message Lengths:
[111 29 155 49 61 147 77 160 157 154]

2. Total Sum of Message Lengths:
447816

3. Average Message Length:
80.37

4. Maximum Message Length:
910

5. Minimum Message Length:
2

6. Median Word Count:
12.0

7. Std Dev of Message Lengths:
59.9216

8. Number of Spam Messages (label=1):
747

9. Top 10 Message Lengths (Descending):
[910 790 629 611 588 588 482 461 458 446]

10. Index of the Longest Message:
1085
```

Output of Pandas Operations:

11. First 5 rows:

Category msg_length

word_count ham 111 20 ham 29

6 spam 155 28 ham 49 11 ham

61 13

12. Last 5 rows:

Category msg_length

word_count spam 160 30 ham 36

8 ham 57 10 ham 125 26 ham 26

6

13. Column Names and Data Types:

Category str Message

str label int64

msg_length int64

word_count int64

char_per_word

float64

14. Dataset Shape (rows, columns):

(5572, 6)

15. Average Message Length by Category:

Category

ham 71.45

spam

137.99

```
16.      Total Messages per Category:
Category
ham 4825
spam 747

17.      Average Word Count by Category:
Category
ham 14.31
spam 23.81

18.      Top 5 Longest Messages:
Category msg_length
word_count ham 910 171 ham
790 162 ham 629 109 ham 611
103 ham 588 125

19.      Bottom 5 Shortest Messages:
Category msg_length
word_count ham 2 1 ham 2 1
ham 2 1 ham 2 1 ham 2 1

20.      Summary Statistics: msg_length word_count char_per_word label count 5572.00 5572.00
5572.00 5572.00 mean 80.37 15.58 5.17 0.13 std 59.93 11.41 1.13 0.34 min 2.00 1.00 2.00 0.00 25%
35.75 7.00 4.57 0.00
50% 61.00 12.00 5.00 0.00
75% 122.00 23.00 5.58 0.00
max 910.00 171.00 49.00
1.00
```

Thank you